

**Zonal asymmetries, teleconnections, and annular
modes in a GCM**

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Abstract

The influence of zonally asymmetric boundary conditions on the leading modes of variability in a suite of atmospheric general circulation models is investigated. The set of experiments consists of nine model configurations, with varying degrees of zonal asymmetry in their boundary conditions. For each configuration, the structure of the leading EOF of the surface pressure is found to vary strongly with the zonal asymmetry of the base state. In particular, a close relationship is found between the structure of the EOF and the model stormtracks. An approximately linear relationship is found to hold between the magnitude of the zonal asymmetry of the leading EOF and of the stormtracks in the models. It is shown that this linear relationship extends to the observations.

One-point correlation maps are calculated for each point in the model domains. Maps taken from the regions where the EOFs reach their maximum amplitude show similar structures for all configurations. These structures consist of a north-south dipole, resembling the observed structure of the NAO. They are significantly more zonally localized than the leading EOF, but do resemble one-point correlation maps and sector EOFs calculated for a simulation with zonally symmetric boundary conditions. Thus, the leading EOF for each simulation appears to represent the longitudinal distribution of NAO-like teleconnection patterns. This longitudinal distribution is strongly influenced by the structure of the stormtracks, both in the models and in the observations.

1 Introduction

A common view of atmospheric variability, dating to the work of Walker and Bliss (1932), is that low-frequency variability is dominated by regional patterns, with scales that are greater than synoptic but significantly smaller than the full hemisphere. These patterns were investigated quantitatively in Wallace and Gutzler (1981) through correlation maps, and are commonly referred to as teleconnection patterns. In the intervening years, numerous studies (e.g., Dole and Gordon 1983; Barnston and Livezey 1987; Hurrell 1995, 1996; Feldstein 1998) have considered the dynamical mechanisms that underlie these features, as well as their impacts on temperature, precipitation, and other quantities of interest. One of the most prominent of these teleconnection patterns is the North Atlantic Oscillation (NAO). The NAO consists of a dipole pattern in sea-level pressure, among other fields, with nodes in the Atlantic basin over Iceland and the Azores. The NAO has long been considered a dominant pattern of low-frequency variability (here defined as periods greater than 10 days) during the Northern Hemisphere (NH) winter, and significantly influences temperature and precipitation over much of the North Atlantic. Thus, an understanding of this feature is of substantial interest from the standpoint of both practice and theory.

In recent years, the suggestion has been made that the observed low-frequency variability can be better explained by hemispheric-scale, zonally symmetric,

rather than asymmetric, modes of variability (see Baldwin and Dunkerton 1999; Gong and Wang 1999; Thompson and Wallace 1998, 2000; Limpasuvan and Hartmann 2000). This alternate hypothesis has generated a certain amount of discussion and controversy in the literature. In particular, it has been argued (e.g., Wallace 2000) that the Northern Hemisphere annular mode (NAM) and the NAO are in fact two means of describing the same phenomenon. However, considering the NAO to be a regional manifestation of a hemispheric-scale pattern leads to a rather different view of Northern Hemisphere variability than does considering the NAO to be a leading mode in its own right.

In a companion paper to the work presented here, Cash et al. (2002) (hereafter CKV) examine the annular modes of an idealized, zonally homogeneous (i.e., the statistics of the model are zonally uniform), GCM. The primary purpose of that study is to address whether or not the model annular mode is representative of the underlying variability of the model. While the annular mode of the zonally homogeneous model is, for sufficiently long integrations, zonally symmetric, and in general consistent with the observations, it is not representative of the underlying variability of the model. Rather, the authors find that the dominant patterns of low-frequency variability are zonally localized dipoles, with structures similar to that of the NAO. In particular, individual days with strong projections onto the annular mode (also referred to as annular mode events) display structures which do not resemble the structure of the annular mode itself; rather, they more resemble the NAO.

In the current study, we extend the work of CKV to include a suite of idealized models, with varying degrees of zonal asymmetry in the boundary conditions. By comparing results from the zonally homogeneous and inhomogeneous models, we can quantify the impact of lower boundary zonal asymmetries on the leading modes of variability in a simplified setting. In particular, we are interested in understanding how both the leading EOF and the teleconnection patterns evolve as the zonal asymmetry of the model base state increases. By considering idealized models, we examine the impact of specific forms of zonal asymmetry in the absence of some of the complex, nonlinear interactions that characterize the climate system. While the idealized nature of our model configuration makes direct comparison to the observations difficult, they should prove sufficiently similar to provide insight into the observations.

The paper is organized as follows. In section 2, we describe the base model used for all simulations discussed here, as well as the details of the individual runs that make up the suite of experiments. The base states from three models, chosen to represent the range of asymmetries in the study, are also considered in detail. In section 3, we present the leading EOFs for the same three models, and describe the relationship between the EOFs and the model base states. In section 4, we compare the leading EOFs and the teleconnection patterns. Finally, section 5 presents a summary and our conclusions.

2 Models and Simulation Characteristics

2.1 Model configurations

The simulations examined in this study are based on the same model used for the zonally homogeneous simulations in CKV, to which readers are referred for a more detailed description. Briefly, the model is a spectral AGCM with T42 horizontal resolution and 14 sigma levels vertical resolution, coupled to a 40 m slab mixed-layer. The model includes full radiative and moisture packages, and is forced by seasonally varying insolation. Each simulation is run for a total of 30 years, with analysis performed on the last 20 years. All analyses presented here were performed for the winter season (December-January-February in the NH, June-July-August in the SH). However, as all of the models have identical boundary conditions for the northern and southern hemispheres, winter seasons for the two hemispheres are combined for each run to produce a 40 year record.

The experiments performed in this study consist of introducing zonal asymmetries of varying strengths and configurations into the boundary conditions of the base model. These asymmetries are introduced in two ways. In one method, the depth of the mixed-layer is reduced for half of the domain (see Fig. 1a for configuration). This shallow region is used to simulate land. The depth of the shallow region, relative to the base depth of 40 m, determines the strength of the zonal surface temperature gradient in the presence of sea-

sonally varying insolation, and hence the magnitude of the zonal asymmetry in the simulation base state. Only the depth of the shallow layer is varied from run to run, the horizontal dimensions being fixed. Our other method of introducing asymmetries in the model is the addition of a gaussian mountain, with a varying maximum height. The horizontal scale is chosen to be roughly equivalent to that of the Tibetan Plateau, and is meant to be a simplified representation of this feature (see Figs. 1b,c). Similar to the asymmetries introduced through changing the mixed-layer depth, only the height of the mountain varies between runs. In total, nine model configurations have been analyzed, using various combinations of mixed-layer depths and mountain heights (see Table 1 for summary).

In this study, we present a detailed analysis of three of these experiments, which span the range of boundary asymmetries. In all discussions of the model configurations, "land" refers to a shallow region of the mixed-layer (i.e., depths less than 40 m), "mountain" refers to the gaussian mountain, and "ocean" refers to regions of 40 m mixed-layer depths.

2.2 Description of model base states

In the first model configuration examined in this study, referred to as L, (see table 1) the only zonal asymmetry introduced consists of reducing the mixed layer depth to 10 cm in the land portion of the domain (see Fig. 1a). The resulting zonal gradient in heat capacity produces a strong seasonal cycle in the

zonal temperature, with a magnitude similar to that seen in the observations (Peixoto and Oort 1992).

In the second model configuration considered, referred to as **H**, we introduce a gaussian mountain with a maximum height of 2500 m, and a mixed-layer depth of 20 m in the land portion of the domain (Fig. 1b). As we would expect, the impact of the zonal asymmetries in case **H** on the base state temperature is far weaker than in the land only case, although it is not negligible.

Finally, we consider the combined impact of land and topography, a configuration referred to as **C**, by including both land (with a depth of 10cm) and the gaussian mountain discussed above, now with a maximum height of 5000m. (Fig. 1c). Note that the mountain is not centered over the land, but is instead at roughly the same remove from the east coast of the land as Tibet is from the eastern Asian coast. Zonal temperature contrasts range from 10 to 20 degrees C in the mid-latitude to polar regions, again in rough agreement with observed Northern Hemisphere winter values.

As we would expect, the different model configurations result in significantly different mean states in all variables, not only surface temperature. In the **L** case (Fig. 2a), we see that the sea-level pressure (SLP) in the NH is dominated by a cold high over land near 30 degrees north, and a warm low over the ocean near 60 degrees north. Consistent with observations, we find summer subtropical highs over the oceans. The horizontal extent of these features is clearly controlled by the width of the land and ocean regions.

In the **H** case, the primary zonal asymmetries in SLP lie downstream of the mountain (Fig. 2b), with relatively low-pressure at high-latitudes and higher pressures towards the pole. The zonal asymmetries in the land region are reduced relative to the **L** run, consistent with the reduction in zonal temperature anomaly. The **C** run shows elements of both the **L** and **H** runs (Fig. 2c), with strong zonal asymmetries in both the land and ocean regions of the domain.

The influence of the different boundary conditions is also clearly visible in the upper levels of the model. In the **L** case, the zonal jet at 200 mb (Fig. 3a) begins near the downstream edge of the land at 35 degrees north latitude. The jet extends for effectively the length of the ocean basin, and shows little meridional tilt. In contrast, the jet in the **H** case (Fig. 3b) is closely confined downstream of the mountain, and shows a distinct south-west north-east tilt. The contrast between the northern and southern hemispheres is reduced relative to case **L**, consistent with the greater thermal inertia of the 20m mixed-layer depth in the land region. In the **C** run, the jet is more sharply localized than in either of the other runs, reaching its maximum value just off the downstream edge of the continent. The position and magnitude of the jet relative to the terrain features is in qualitative agreement with the observed position and intensity of the NH wintertime Pacific jet.

The differences between the three model runs are particularly clear in their stationary wave patterns. The **L** run (Fig. 4a) displays a relatively simple wavenumber-1 pattern in the midlatitudes, with lowered heights over the ocean

region. In the **H** run (Fig. 4b) the most significant feature is a small region of lowered heights to the north-east of the mountain. As with the upper-level jet, the **C** run (Fig. 4c) displays features related to both the **L** and **H** run.

In summary, the three configurations of the model boundary conditions result in zonal asymmetries that are distinct for each run, from the weakly zonally inhomogeneous **L** run to the fairly realistic **C** run. This makes them suitable for examining the impact of varying asymmetries on the structure of the leading EOFs.

3 Annular Modes

As a means of characterizing the low-frequency variability of each model, we calculate the leading EOF of the surface pressure. For this calculation, we first remove the seasonal cycle, by subtracting the calendar mean of the surface pressure at each day. The detrended data is then 10-day low-pass filtered, and area weighted by the square root of the cosine of the latitude. This filtered, weighted data is then used to calculate the leading EOF for each model (Fig. 5).

In the meridional direction, the structure of the leading EOF is similar for each model over most of the domain. Each EOF consists of a dipole, with maxima in the polar region and the midlatitudes, and the node falling near 60° N latitude. However, the zonal structure of the modes clearly varies with

the zonal inhomogeneity of the base state of the model. In the **L** case, positive loadings extend around the entire hemisphere (Fig. 5a), with the maximum midlatitude values of the EOF off the east coast of the land region. In the **H** case (Fig. 5b), the EOF is more localized over the ocean, and the EOF maximum is again located downstream of the terrain feature. The **C** run (Fig. 5c) EOF is even more localized over the ocean, and the location of the maximum is similar to the previous runs.

Comparing the structure of the leading EOFs to the model base states, we find that for each model the leading EOF takes on its maximum amplitude in the midlatitudes, slightly downstream of the zonal jet (compare Fig. 3 and Fig. 5). This suggests that processes in the jet exit region may be playing a role in determining the position and amplitude of the leading EOF. To investigate this more directly, we compare the position of the model stormtracks and leading EOFs (Fig. 6). Here we define the stormtrack as the time-mean 2-10 day bandpass-filtered eddy kinetic energy, vertically averaged from 1000 to 100 mb. In general, the zonal variations in the leading EOF are closely related to the zonal variations in the model stormtrack.

This relationship is illustrated by comparing the deviations from zonal symmetry in the leading EOF to the deviations from zonal symmetry in the stormtrack (Fig. 7) for *all* the model configurations listed in Table 1. As is apparent from Fig. 7, an approximately linear relationship exists between the two quantities for moderate values of the asymmetry, with a hint of a saturation of the

asymmetry of the eddy kinetic energy. It is of particular interest to note that this relationship is not limited to the models alone. Also plotted in Fig. 7 are values for the Northern and Southern Hemisphere winter seasons, as determined from the NCEP/NCAR Reanalysis data. Although the data point for Northern Hemisphere indicates a somewhat higher value of EOF asymmetry, the asymmetry of the Northern and Southern Hemisphere EOFs bears a similar relationship to the asymmetry of the corresponding eddy kinetic energy as is seen in the idealized models. It should be noted that the relationship between the stormtracks and the leading EOF in the models and observations, while similar, is not exactly the same. In the observations the zonal anomalies in the leading EOFs tend to lie a little downstream of the stormtrack zonal anomalies. This is not found in all model simulations — in some integrations the deviations from the zonal mean in stormtracks and the EOFs tend to nearly coincide, although in others the EOF is a little downstream of the eddy kinetic energy maximum. However, it appears that a robust relationship does exist between the stormtracks and the leading EOFs, in both the models and the observations. Specifically, the zonal asymmetries of the leading EOFs appear to be closely tied to the zonal asymmetries of the stormtracks.

4 Teleconnection patterns

We have seen that there is relationship between the leading EOFs and the stormtracks; in this section we examine further the physical significance of that relationship. One issue is that the leading EOF may not be representative of the underlying patterns of variability (CKV; Ambaum et al. 2001). In the observations and the zonally homogeneous model, one-point correlation maps (also referred to as teleconnection patterns) reveal that the dominant patterns of low-frequency variability are zonally localized, with characteristic zonal scales of approximately 60° longitude. In CKV, we show that these teleconnection patterns are also similar in structure to the leading EOF of the model calculated for a limited sector (see Fig. 8) (in having a meridionally-oriented dipole) indicating that these localized dipoles represent a prevalent pattern of low-frequency variability in the aquaplanet model.

To investigate the relationship between the leading EOFs and the underlying data in the zonally inhomogeneous models, we calculate one-point correlation maps for each point in the model domain. The strength of the teleconnection pattern for each base point is defined as the absolute value of maximum anticorrelation that is found, anywhere in the domain, for that base point. Examining the geographical distribution of the maximum anticorrelations (Fig. 9), we find that the maximum anticorrelations tend to cluster in those regions of the model domain where the leading EOF has high amplitudes as well. We

also find that the minima in the anticorrelations tend to fall along the zero lines of the EOFs. In the **L** and **H** configurations of the model (Figs. 9a,c), we see that the highest magnitudes of the teleconnection patterns are closely associated with the maxima of the leading EOF. In the **C** configuration, there are two distinct regions with high-amplitude anticorrelations. One region, lying near the position of the mountain (160° W longitude) is associated with the presence of the mountain itself, and does not show any particular association with the EOF. The second region of strong teleconnections does appear to be associated with the maximum in the low-latitude center of the EOF. The general relationship between regions of strong teleconnections and high leading EOF magnitudes also holds in the other model integrations performed (not shown). In the zonally homogeneous model used in CKV, we also found similar meridional structures for the maxima and minima in the one-point correlations and the leading EOF of the zonal-mean surface pressure.

Having established that a similar relationship exists between the teleconnections in the zonally homogeneous and inhomogeneous models, we now consider the structure of those teleconnections. Similar to the analysis of Wallace and Gutzler (1981), we examine the one-point correlation map for the base point with the maximum anticorrelation, for each of the three simulations considered here (Fig. 10a-c). Although there are some variations between individual models, the general pattern is clearly that of a dipole, with the low-latitude center falling near 40° latitude, and a higher-latitude center near 80° . Of partic-

ular interest is the similarity of the teleconnection patterns derived for the zonally inhomogeneous models to that calculated for the CKV aquaplanet model (compare to Fig. 10d). The centers of the teleconnections are also offset in the east-west direction by about 20° longitude in each configuration, similar to the sector EOF.

The similarity between the teleconnections and the sector EOF from the zonally homogeneous model, combined with the overlap between the regions of strong teleconnections and the leading EOFs, suggests that the annular mode represents the averaged longitudinal distribution of individual events that have a structure similar to that of the teleconnections. To test this hypothesis, we calculate a measure of the average pattern correlation between the sector EOF (Fig. 8) and the low-pass filtered surface pressure as a function of longitude for each of the model configurations. The methodology of this test is as follows. For each configuration, we first center the sector EOF at a given longitude. We then calculate the time-mean of the absolute value of the pattern correlation between the sector EOF and the low-pass filtered surface pressure in the sector of the model corresponding to the location and dimensions of the EOF. The resulting pattern correlation magnitude is recorded for the chosen central longitude, and the process is then repeated for each longitude in turn.

For each run, the resulting correlation plots show good agreement with the structure of the respective EOFs (Fig. 11). Correlations reach their peak values downstream of the terrain features, and in general follow the amplitude of

the midlatitude center of the EOF. This supports the hypothesis that individual events have structures similar to the sector EOF, and that the hemispheric EOF represents the distribution of these events. In general, it appears that the leading EOF for each model can be thought of as reflecting the distribution of the dipole patterns reflected by the sector EOF. The fact that the agreement is in general fairly close is remarkable, given that the sector EOF is derived from a separate model, and is a testament to the robustness of the localized dipole.

Given the similarity between the model teleconnections and the NAO, and the fact that regions of high teleconnection strength tend to correspond to high amplitudes of the leading EOF, a number of features of the model can now be explained. Numerous studies have linked the existence and behavior of the NAO to the stormtracks (e.g., Rogers 1990; Hurrell 1995; Gulev 1997; DeWeaver and Nigam 2000). While the relationship between the stormtracks and the NAO is complex, and not fully understood, evidently the high-frequency transients play an important role in governing the NAO.

Similar to the observations, the most prominent teleconnections in our models occur near the stormtracks. Thus, it appears that the relationship between the leading EOFs and the stormtracks in our models comes about through the link between the high-frequency transients and the NAO-like teleconnection patterns. Consistent with the leading EOFs/annular modes of the zonally homogeneous model in CKV, the leading EOFs/annular modes of the zonally inhomogeneous models represent the longitudinal distribution of the dominant

teleconnection patterns, rather than representing a single, dynamical feature. Unlike CKV, the teleconnection patterns are now distributed inhomogeneously in longitude. The characteristics of that distribution appear to be governed by the stormtracks.

To further investigate the relationship between the teleconnections and the high-frequency transients, we define a teleconnection index for each model configuration. Similar to the NAO index used by Wallace and Gutzler (1981), we define the teleconnection index for a given model configuration as the difference between the surface pressure at the high-latitude and low-latitude centers of the teleconnection pattern. Thus, to take the C model configuration as an example, the index is defined as

$$I(t) = P_s(75^\circ N, 100^\circ W) - P_s(45^\circ N, 10^\circ W) \quad (1)$$

where P_s is the surface pressure. The index is then normalized by its standard deviation, and lag-regressed against the quantities of interest.

We now assess, in a simple way, the role of the high-frequency transient eddy forcing in the teleconnection events. In Fig. 12, we compare the zonal wind anomaly (shading) associated with each event to the meridional contribution of the high-frequency eddies to the eddy momentum flux convergence (contours). The lag-regressed zonal wind anomaly and eddy forcing are zonally averaged over a 90° longitude region centered on the teleconnection. In both the zonally inhomogeneous (Figs. 12a-c) and the zonally homogeneous

(Fig. 12d) configurations, we see a dipole pattern in the zonal wind anomalies, associated with a shift in the maximum of the zonal jet. Similarly, we find a dipole pattern of the same sign in the eddy forcing, which tends to lead the dipole in the zonal winds on the order of a few days. It is particularly noteworthy that the results for the zonally homogeneous and inhomogeneous models are nearly identical, strengthening the view that the dynamical processes governing the dominant patterns of low-frequency variability are similar in each case, and only the preferred location of occurrence changes between models.

5 Summary and Conclusions

In this study, we have considered the leading patterns of variability in a suite of idealized GCMs. For each model configuration, different asymmetries were introduced in the boundary conditions, and the impact of these asymmetries on the model annular modes was investigated. We find that a robust relationship exists between the zonal inhomogeneity of the model base state and the zonal asymmetry of the leading EOF. In particular, the leading EOF of the surface pressure tends to follow the structure of the model stormtrack, as defined by the vertically averaged bandpass-filtered eddy kinetic energy. As the zonal asymmetry of the model base state increases, the stormtracks and leading EOFs become increasingly zonally localized. Consistently, the regions in which we find strong teleconnection patterns also become more zonally local-

ized. We find a near-linear relationship between the rms deviation from the zonal mean of the EOF and the stormtrack. Of course, one must be cautious in making any direct comparisons between our model and observations, because of the highly idealized nature of the simulations performed.

As in CKV and Ambaum et al. (2001), we find that the leading EOF does not resemble the zonal covariance structure of the variability. One-point correlation maps consistently demonstrate that the model intra-seasonal variability (i.e., on the weekly to monthly timescale) tends to be dominated by more localized patterns, with characteristic zonal scales of approximately 60° longitude. Of particular interest is the fact that the scale and structure of the teleconnection patterns are similar between model runs, despite the significant differences in boundary conditions. These differences in boundary conditions appear to manifest themselves primarily in the longitudinal distribution of occurrence of the teleconnections, rather than in the structure of the patterns themselves. We also find that the regions of strongest teleconnections generally follow the pattern of the leading EOF, although the match is not exact in all cases.

The above results lead to the conclusion that the leading EOF of each model generally represents the longitudinal distribution of meridional dipole patterns within the model domain. These patterns resemble the observed NAO, and there is evidence to suggest that this relationship holds in the atmosphere as well. During the NH hemisphere winter, the Atlantic stormtrack is substan-

tially stronger than the Pacific stormtrack (due, at least in part, to the so-called “mid-winter suppression” phenomenon). As is well known, the dipole pattern of the NAO is the dominant pattern of low-frequency variability in the North Atlantic during the winter months. In the North Pacific lesser known, but still prominent dipole teleconnection pattern (the North Pacific Oscillation, or NPO) is also observed during the winter (Wallace and Gutzler 1981). Due to the more complex geometry of the NH, patterns such as the Pacific-North American teleconnection pattern are prominent in the observations, but absent from the models considered here. However, the presence of the NAO and the NPO, combined with their relative strength and that of the stormtracks in the respective ocean basins, suggests that the dipole patterns observed in the models represent a physical manifestation of intra-seasonal variability. For more zonally symmetric geometry, the storm tracks are correspondingly more zonally symmetric and this leads to a more zonally symmetric first EOF, an annular mode. In this picture, the mechanisms producing the NAO and the annular mode are the same, namely the forcing by baroclinic instability in storm tracks. It is then the meridional dipolar structure of the pressure field, which is in turn related to the variations in the zonal wind, that is the robust physically meaningful structure and that is the building block of both phenomena.

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		Mountain Height		
		0	2500	5000
Land	10cm	L	1	C
Depth	20m	2	H	3
	40m	A	4	5

Table 1: Experimental design. The above table lists the mountain heights and mixed-layer depths used to simulate land in each of the experiments considered in this study. For all cases, the shallow mixed-layer region representing land extends from 90° S to 90° N latitude, and from 70° E to 110° W longitude. The gaussian mountain is centered at 140° W longitude and 35° latitude in each hemisphere. Half-width is 15° in both longitude and latitude. Experiment A refers to an aquaplanet, with a 40m mixed layer ocean everywhere and no distinguishable land. Experiment L has a (saturated) land mass of small heat capacity but no mountain. Experiment H has a mid-size mountain and a mild contrast between ‘land’ and ocean. Experiment C has land mass and a mountain, and is the most zonally asymmetric of all. The other configurations are combinations of these, and are referred to by number.

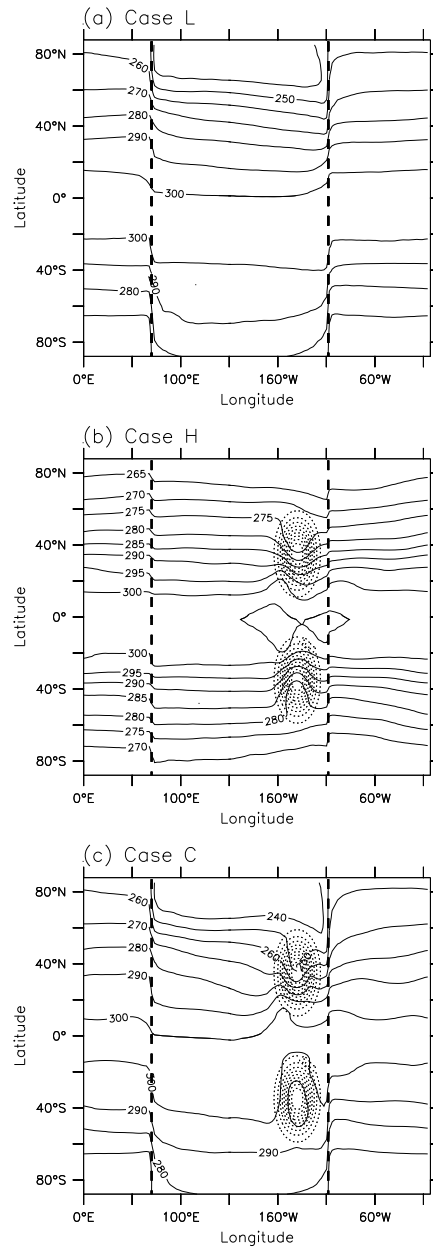


Figure 1: NH winter mean surface temperatures for (a) L case, (b) H case, and (c) C case. Heavy contours are surface temperature (degrees K), dashed lines denote edge of 10cm mixed-layer representing land, and dotted lines the gaussian mountain (maximum height is 5000 m).

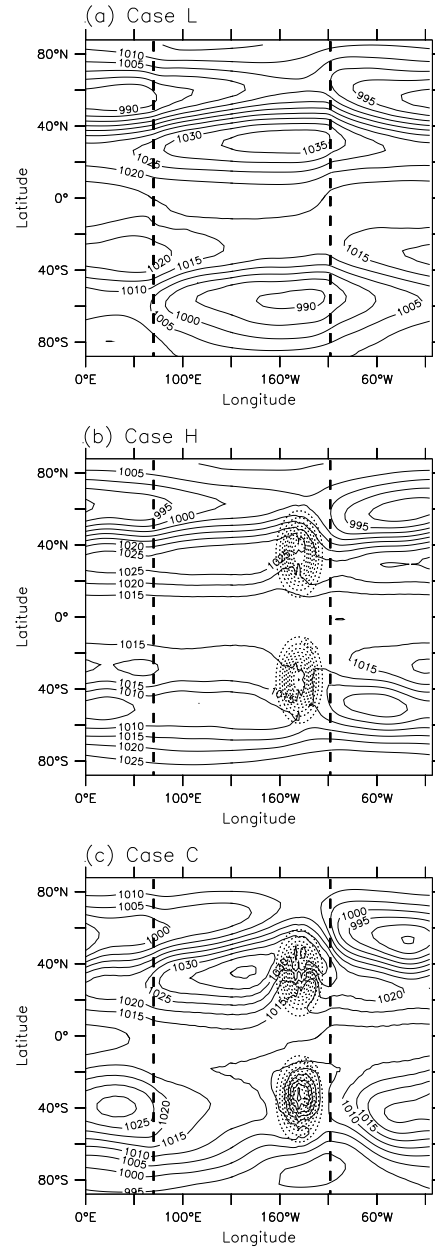


Figure 2: NH winter mean sea-level pressure for (a) L case, (b) H case, and (c) C case. Heavy contours are sea-level pressure, dashed lines denote edge of 10cm mixed-layer representing land, and dotted lines the gaussian mountain (maximum height is 5000 m).

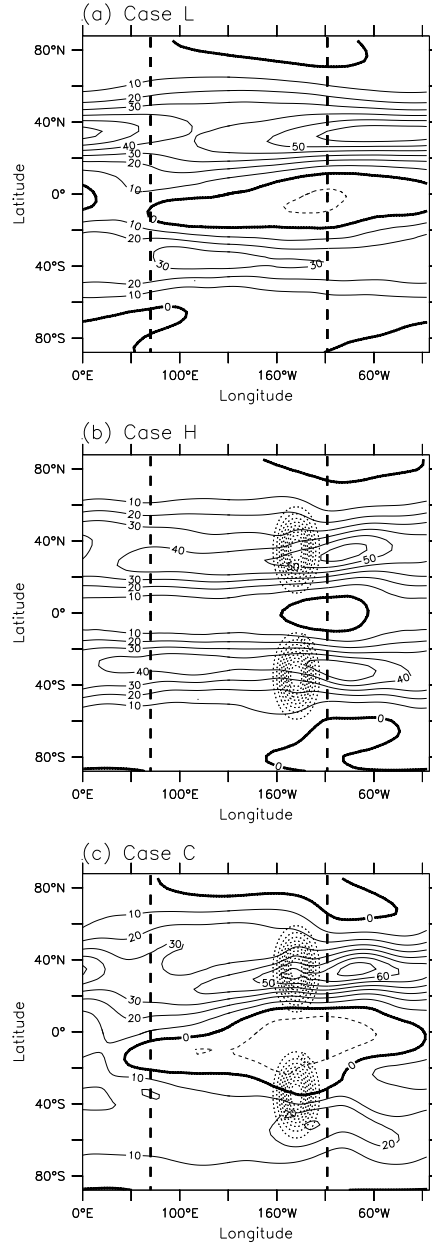


Figure 3: NH winter mean 200 mb zonal winds for (a) L case, (b) H case, and (c) C case. Heavy contours are 200 mb zonal wind, dashed lines denote edge of 10cm mixed-layer representing land, and dotted lines the gaussian mountain (maximum height is 5000 m).

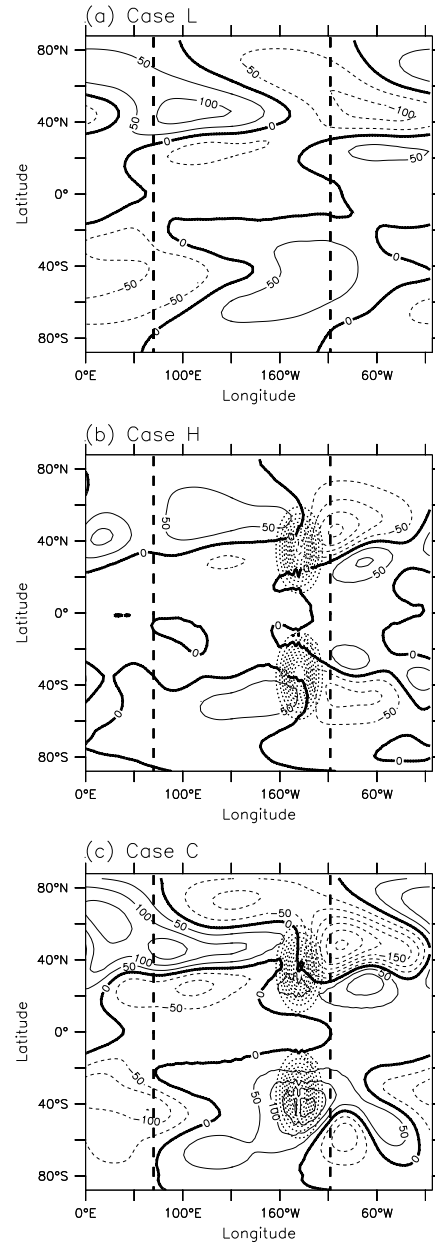


Figure 4: NH winter mean 200 mb stationary waves for (a) L case, (b) H case, and (c) C case. Stationary waves are defined here as deviations of the winter mean 200 mb geopotential heights from their winter mean zonal mean. Heavy contours denote geopotential height anomaly. Solid lines are positive, dashed lines are negative. Other features are as given in Fig. 3.

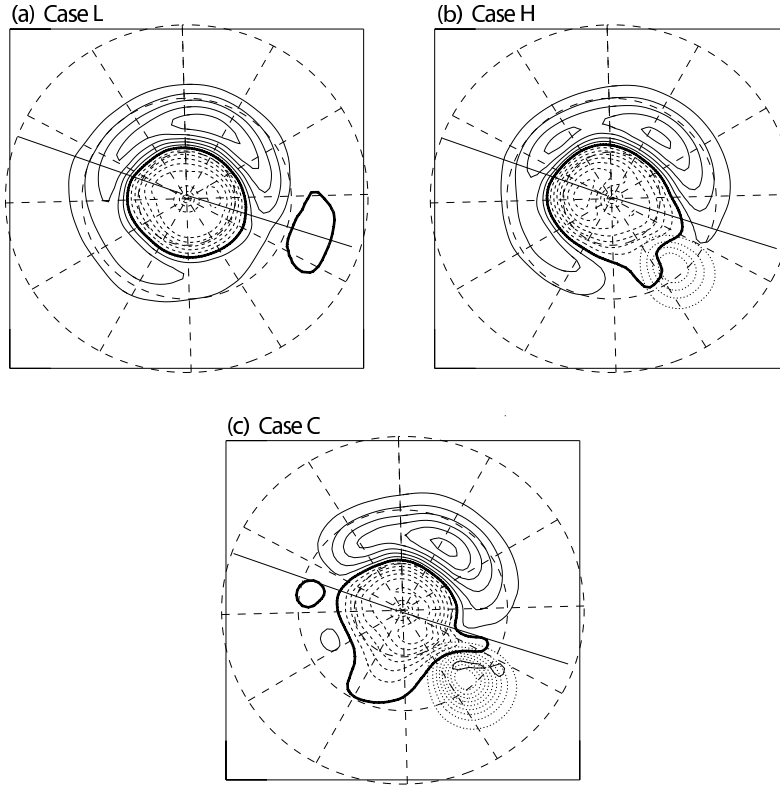


Figure 5: Annular mode for (a) **L** case, (b) **H** case, and (c) **C** case. Annular modes are defined as the leading EOF of the surface pressure for each model. Solid contours are positive, dashed lines are negative. Other features are as given in Fig. 3 (note land region lies towards the bottom of each panel).

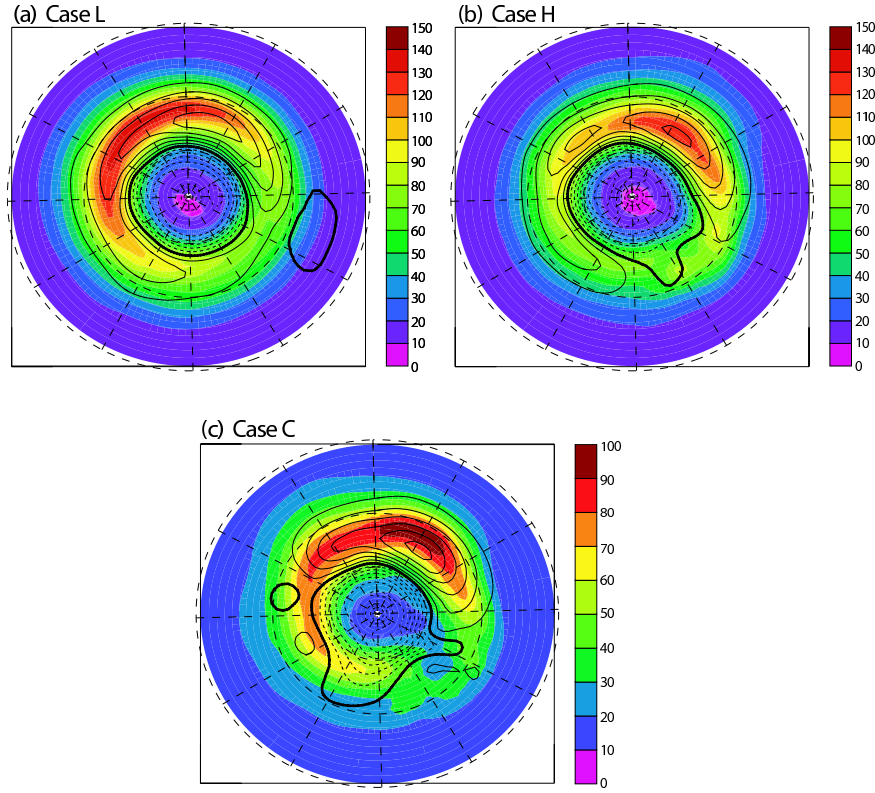


Figure 6: Comparison of leading EOF and stormtrack for (a) **L** case, (b) **H** case, and (c) **C** case. Shading denotes model stormtrack, as defined by the vertically averaged 2-10 bandpass filtered eddy kinetic energy. Contours denote the model leading EOF. Solid contours are positive, dashed lines are negative. Geographical features have been omitted for clarity.

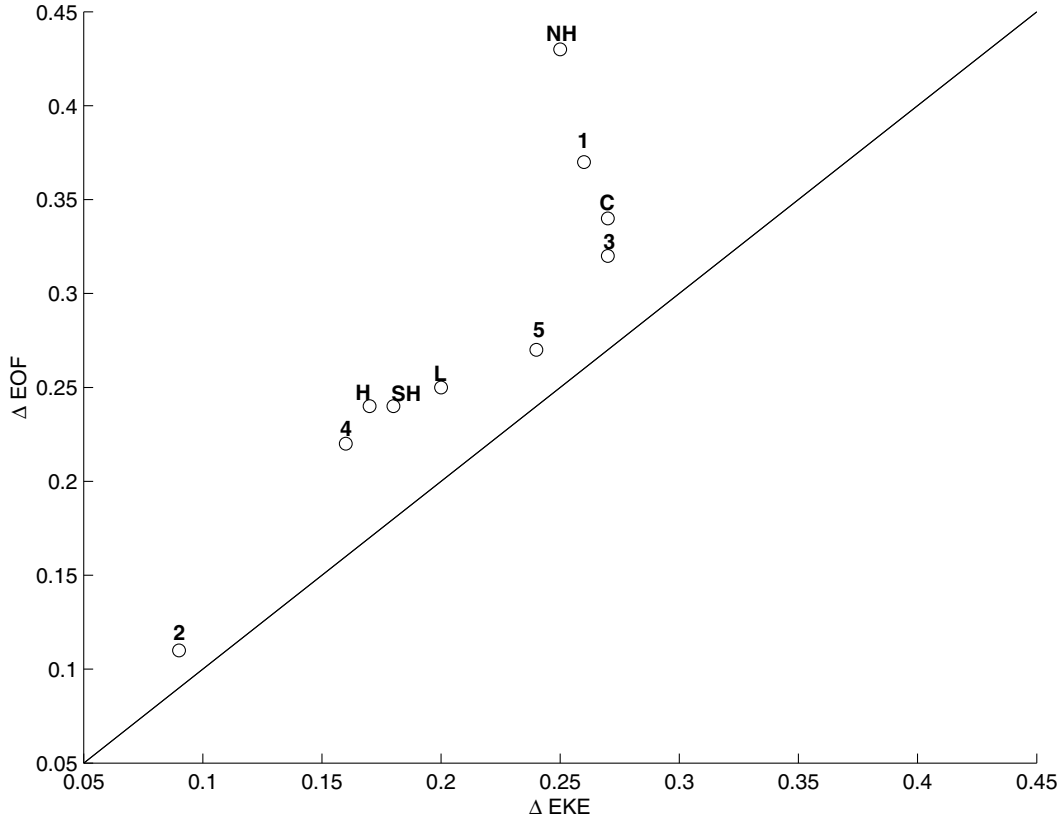
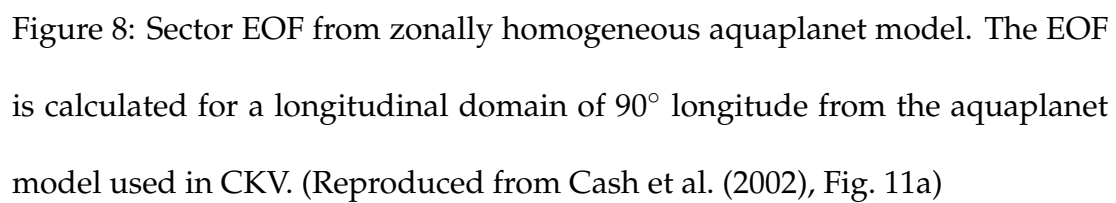


Figure 7: Comparison of zonal asymmetry of leading EOF with zonal asymmetry of stormtrack. Zonal asymmetry is defined as rms deviation from zonal mean for both quantities, normalized by zonal mean values. Labels are as given in Table 1, and NH and SH refer to Northern and Southern Hemisphere winters, respectively.



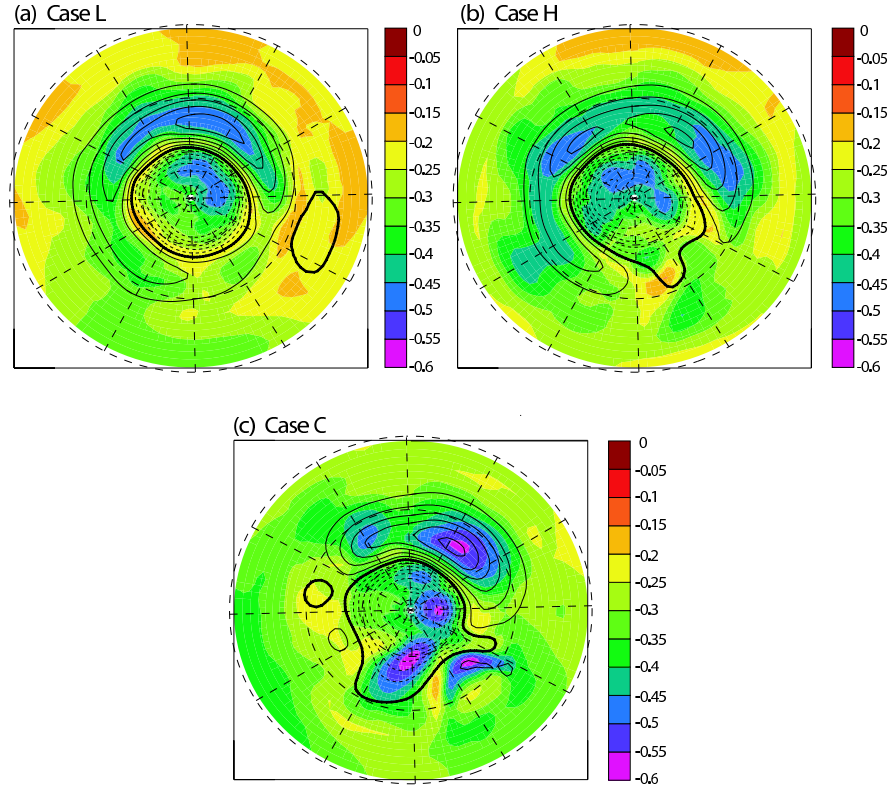


Figure 9: Comparison of maximum anticorrelation and leading EOF for (a) L case, (b) H case, and (c) C case. Shading denotes the value of the maximum anticorrelation for a given basepoint, and contours denote the model leading EOF. Solid contours are positive, dashed lines are negative. Unshaded regions indicate absolute values greater than 0.5. Geographical features have been omitted for clarity.

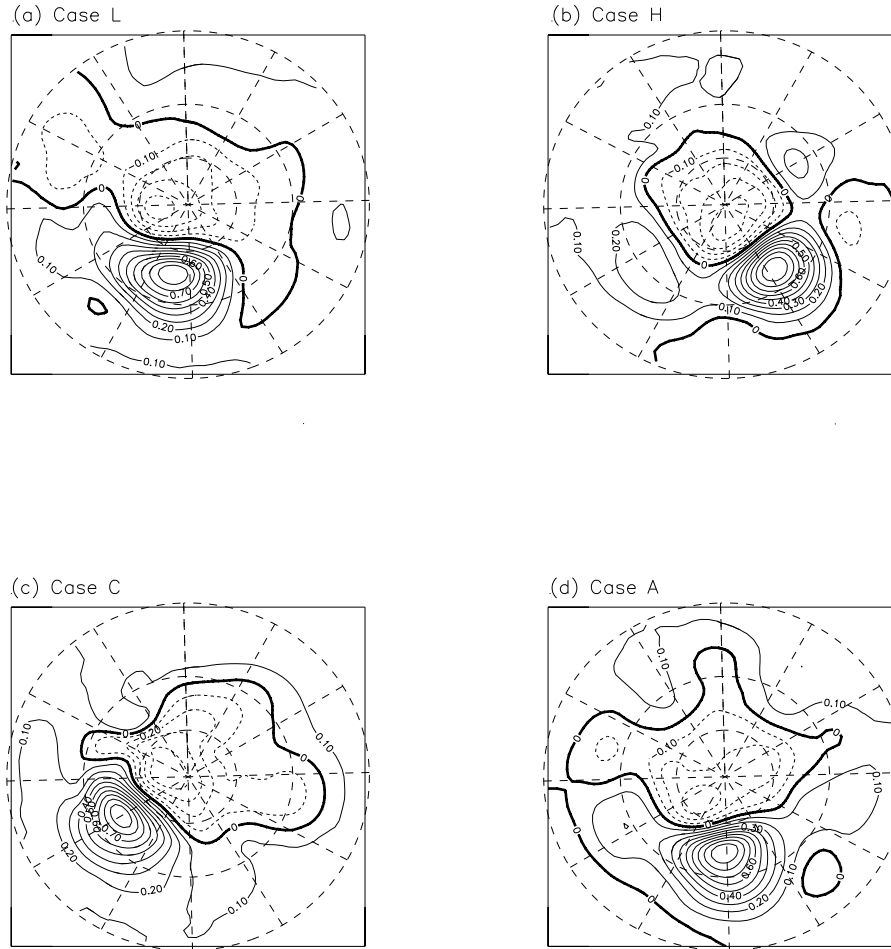


Figure 10: Teleconnection patterns for (a) **L** case, (b) **H** case, (c) **C** case, and (d) **A** case (from CKV). Contours denote one point correlation maps, taken for the basepoint of the maximum anticorrelation for each model. Solid contours are positive, dashed lines are negative, and the contour interval is 0.1. Geographical features have been omitted for clarity.

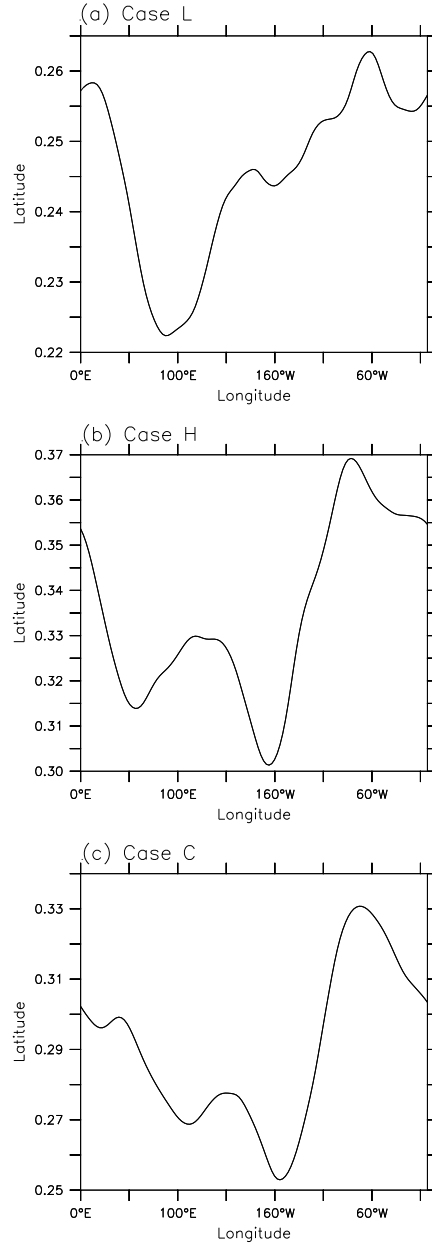


Figure 11: Pattern correlation between low-pass filtered surface pressure from (a) **L** case, (b) **H** case, and (c) **C** case and sector EOF. Values represent pattern correlation with sector EOF shown in Fig. 8, centered at the given longitude.

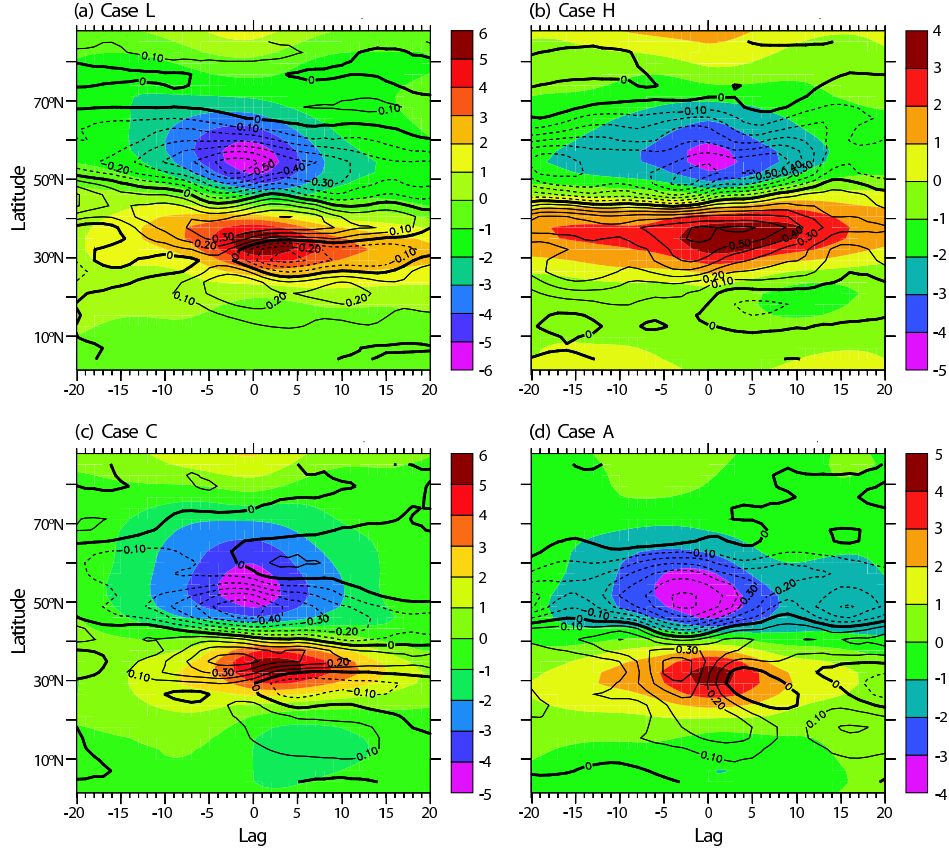


Figure 12: Regressed zonally averaged zonal wind anomalies (shading) and anomalous eddy momentum flux convergence (contours) for (a) case L, (b) case H, (c) case C, and (d) case A. Units of wind are in ms^{-1} , units of momentum flux convergence are in $ms^{-1} day^{-1}$.